Test Build

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1 Where We At

We have a consumption-savings problem with utility $u(c) = \log(c)$ and budget:

$$a_{t+1} = (1 + r)a_t - c_t$$

2 Why We Get Motivated

The value function is:

$$V(a_0) = \sum_{t=0}^{\infty} \beta^t \log(c_t^*)$$

But this solves only for a specific a_0. We want a rule for any wealth level.

3 What We Do

Separate the first term and re-index:

$$\begin{aligned} V(a_0) &= \log(c_0^*) + \beta \sum_{t=1}^{\infty} \beta^{t-1} \log(c_t^*) && (1) \\ &= \log(c_0^*) + \beta V(a_1) && (2) \end{aligned}$$

Define vectors $\boldsymbol{x} \in \mathbb{R}^n$ and matrix $\mathbf{A}$. The expectation $\mathbb{E}[c_t^*]$ and $\mathrm{Var}(a_t)$ are finite. The optimal policy is:

$$c_t^* = \underset{c}{\operatorname{argmax}} \{\log(c) + \beta V((1 + r)a_t - c)\}$$

Bellman Equation

This recursive structure $V(a) = \max_c\{u(c) + \beta V(a')\}$ is the Bellman equation. It reduces an infinite-horizon problem to a single-period optimization.

4 Results and Next Steps

Theorem 4.1. *Under $\beta(1+r) < 1$, the value function V is concave and the policy function $c^*(a)$ is increasing in a.*

Lemma 4.2. *The envelope condition gives $V'(a) = (1+r)u'(c^*(a))$.*

Definition 4.1. A **stationary equilibrium** is a distribution μ over wealth such that μ is invariant under the optimal policy.

- Bellman equation established
- Next: solve via FOC + envelope theorem
- Then: characterize the stationary distribution

Note

This is a test of the mynote tcolorbox style from marcoreport.sty.